

Foundations Of Cybersecurity & Environment Setup

Detailed cybersecurity & ethical hacking internship notes

These notes are organized around the task 1 requirements in the uploaded internship plan: cybersecurity basics, lab environment setup, linux fundamentals, networking basics, cryptography basics, and tool familiarization. The deliverables listed in the plan are also included.

Lab used for this internship work: virtualbox, kali linux as the attacker machine, metasploitable2 as the target, a private host-only network, and wireshark for packet capture.

Study note: the explanations below expand the listed task 1 topics for learning and documentation. Commands and hands-on activities should be performed only in your isolated lab.

1. Cybersecurity basics

1.1 What Is Cybersecurity?

Cybersecurity is the practice of protecting computers, networks, applications, devices and information from unauthorized access, misuse, disruption, modification or destruction. Security is not only about tools; it also involves people, processes, configuration, monitoring and responsible behaviour.

1.2 Cia Triad

Confidentiality: information should be accessible only to authorized people or systems. Examples include access controls, authentication and encryption.

Integrity: information should remain accurate and should not be changed without authorization. Hashes, permissions, logging and integrity checks help protect it.

Availability: systems and information should be accessible when needed. Backups, redundancy, monitoring and recovery planning support availability.

1.3 Threat Types

- **Phishing:** deceptive messages or websites intended to trick users into revealing information or performing an unsafe action.
- **Malware:** malicious software designed to damage, disrupt, spy on or gain unauthorized access to systems.

- **Ddos:** a distributed denial-of-service attack attempts to make a service unavailable by overwhelming it with traffic or requests.
- **Sql injection:** unsafe handling of user input can allow unintended sql statements to affect a database.
- **Brute force:** repeated attempts are made to guess credentials or other secrets.
- **Ransomware:** malware that can restrict access to files or systems and commonly demands payment.

1.4 Attack Vectors

- **Social engineering:** attacks that manipulate people rather than relying only on technical weaknesses.
- **Wireless attacks:** attacks that target wireless communications, configurations or access controls.
- **Insider threats:** risks involving trusted users who intentionally or accidentally misuse access.

1.5 Basic Security Principles

- Use strong authentication and least privilege.
- Keep software and operating systems updated.
- Use backups and test restoration.
- Separate sensitive systems from untrusted networks.
- Monitor logs and investigate unusual activity.
- Use security tools only with authorization.

2. Lab Environment Setup

2.1 Purpose Of The Lab

The internship plan requires a virtual practice environment using virtualbox or vmware, kali linux as the attacker machine, metasploitable2 or dvwa as the target, and a private host-only adapter network. The purpose is to practice security concepts without exposing an intentionally vulnerable target to an external network.

2.2 Components

- **Virtualbox:** virtualization platform used to run the virtual machines.
- **Kali linux:** security-focused linux distribution used as the attacker/testing machine.
- **Metasploitable2:** intentionally vulnerable target used for controlled security practice.

- **Host-only adapter:** creates a private network between the host and selected virtual machines.
- **Wireshark:** captures and analysis network packets.

2.3 Example Network Architecture

In your completed setup, kali uses nat for normal connectivity and a host-only interface for the private lab. Your observed addresses were kali 192.168.56.101 and metasploitable2 192.168.56.102.

Kali (192.168.56.101) → host-only network → metasploitable2 (192.168.56.102)

2.4 Verification

Useful verification steps include checking interfaces with `ip addr`, checking routes with `ip route`, and testing reachability with a ping to the target. A successful ping confirms basic ip connectivity; it does not by itself prove that every service is reachable.

2.5 Wireshark Test Capture

For the task 1 evidence, wireshark can capture the icmp traffic generated when kali pings metasploitable2. Select the host-only interface, start capture, generate the ping, stop the capture, and use the display filter icmp to focus on icmp packets.

3. Linux Fundamentals

3.1 File System Navigation

Command	Purpose	Example
Pwd	Shows the current working directory.	Pwd
Ls	Lists files and directories.	Ls -la
Cd	Changes the current directory.	Cd /etc

3.2 Permissions And Ownership

Chmod changes permissions. Linux permissions commonly describe read (r), write (w), and execute (x) access for the owner, group and others. **Chown** changes file ownership and can also change the group.

Examples: `chmod +x script.sh` adds execute permission; `chown user:group file` changes ownership when performed with appropriate privileges.

3.3 Package Management

Apt is a high-level package management tool used on debian-based systems. Common examples include `sudo apt update` to refresh package information and `sudo apt install wireshark` to install a package. **Dpkg** works directly with debian package files and package database operations.

3.4 Networking Commands

- `Ifconfig` displays network interface information on systems where it is installed.
- `Ip addr` is the modern command for viewing ip addresses and interfaces.
- `Ping` tests reachability using `icmp` or another supported mechanism.
- `Netstat` displays network connections and statistics when installed.
- `Traceroute` helps show the network path toward a destination.

3.5 Linux Practice

For task 1, practice each listed command in kali and understand what information it returns rather than simply memorizing syntax.

4. Networking Basics

4.1 Osi Model

Layer	Name	Simple function
7	Application	Network services used by applications.
6	Presentation	Data representation, formatting and related transformations.
5	Session	Manages communication sessions.
4	Transport	End-to-end delivery; tcp and udp operate here.
3	Network	Logical addressing and routing; ip operates here.
2	Data link	Frames and local network delivery.
1	Physical	Signals and physical transmission.

4.2 TCP/IP Protocol Suite

The tcp/ip model describes the protocols and functions used for internet communication. Important protocols include ip for addressing/routing, tcp and udp for transport, dns for name resolution, and http/https for web communication.

4.3 DNS

The domain name system maps human-readable names such as a domain name to ip addresses and supports service discovery. Dns reduces the need for users to remember numeric ip addresses.

4.4 HTTP AND HTTPS

Http is an application-layer protocol used for web communication. Https is http carried over a secure tls connection, providing protection such as confidentiality and integrity during transport when correctly configured.

4.5 IP addressing and subnetting

An ip address identifies an interface within an ip network. A subnet mask or prefix length identifies which part represents the network. Subnetting divides a larger network into smaller networks, which can improve organization and address management.

4.6 NAT

Network address translation changes address information as traffic crosses a network boundary. In home and lab environments, nat commonly lets private systems share a host's external connectivity.

5. Cryptography Basics

5.1 Why Cryptography Is Used

Cryptography provides techniques for protecting information and communications. It can support confidentiality, integrity, authentication and non-repudiation depending on the algorithm and protocol used.

5.2 Symmetric Encryption

Symmetric encryption uses the same secret key, or related secret key material, for encryption and decryption. It is generally efficient for protecting larger amounts of data, but the communicating parties need a secure way to obtain the secret key.

5.3 Asymmetric Encryption

Asymmetric cryptography uses a public/private key pair. The public key can be shared, while the private key is kept secret. It is useful for key exchange, authentication and digital signatures, although it is generally more computationally expensive than symmetric encryption.

5.4 Hashing

A cryptographic hash converts input data into a fixed-length digest. Hash functions are designed to make it difficult to recover the original input or find practical collisions. The task 1 plan specifically names md5 and sha-256. Md5 is historically important but is not suitable for modern security-sensitive integrity applications where collision resistance is required; sha-256 is a stronger modern choice.

5.5 Digital Certificates

A digital certificate associates an identity with a public key and is normally signed by a certificate authority or another trusted issuer. Certificates are a core part of tls authentication.

5.6 SSL/TLS

Tls protects network communication against eavesdropping and tampering when correctly implemented. It is used by https and many other secure protocols. Ssl is the older predecessor; modern secure deployments use tls.

5.7 Openssl Practice

The task 1 plan asks for hands-on encryption and decryption using openssl. Perform this only with test data in your lab and document the command, input, output and result.

6. Tool Familiarization

6.1 Wireshark

Wireshark is a graphical network protocol analyzer. It captures packets from a selected network interface and lets you inspect protocol fields, addresses and traffic patterns. For your task 1 demonstration, capture icmp traffic from a ping between kali and metasploitable2.

Simple workflow: open wireshark → select eth1 → start capture → run ping -c 4 192.168.56.102 → stop capture → filter with icmp.

6.2 NMAP

Nmap is a network discovery and security auditing tool. It can identify live hosts and enumerate ports and services. In this internship, use it only against your authorized lab target.

Basic lab example: nmap 192.168.56.102. Service detection can be performed with nmap -sv 192.168.56.102 when that activity is required by the task.

6.3 Burp Suite

Burp suite is a web security testing platform. Its proxy can sit between a browser and a web application so that requests and responses can be inspected in an authorized test environment.

6.4 NETCAT

Netcat is a network utility that can create tcp or udp connections and is useful for debugging and understanding network services. Use it only between systems you own or are explicitly authorized to test.

7. Task 1 practical evidence & recording

7.1 lab setup screenshots

- Virtualbox showing kali linux and metasploitable2.
- Kali terminal showing network interfaces and the host-only address.
- Metasploitable2 showing its host-only address.
- Successful ping from kali to metasploitable2.
- Wireshark showing the icmp test capture.

7.2 five-minute video order

1. Introduction → 2. Show virtualbox and both vms → 3. Show kali network configuration → 4. Show metasploitable2 → 5. Ping the target → 6. Open wireshark and capture on eth1 → 7. Generate the ping → 8. Show the icmp packets → 9. Give a short conclusion.

7.3 what to say

A simple explanation is: “kali linux is my attacker machine and metasploitable2 is my target. Both are connected through a private host-only network. I verified connectivity with ping and used wireshark to capture the icmp packets generated by the test.”

7.4 safety

Keep the intentionally vulnerable target isolated from untrusted networks. Perform scanning, enumeration and security testing only on systems you own or have explicit authorization to test.